

Chapter - (2)

HCF and LCM of Numbers

Factors and Multiples : $\rightarrow$

If a number ~~a~~ a divides another number b i.e. $\left(\frac{b}{a}\right)$ exactly, we say that a is a factor of b .

In this case, b is called a multiple of a .

Highest common factor (HCF) or

Greatest common Measure (GCM) or

Greatest common Divisor (GCD) : $\rightarrow$

The HCF of two or more than two numbers is the greatest number that divides each of them exactly.

There are two methods of finding the HCF of a given set of numbers :-

Factorization method :- Express each one of the given numbers as the product of prime factors. The product of least powers of common prime factors gives HCF.

Ex. Find the HCF of $2^3 \times 3^2 \times 5 \times 7^4$,
 $2^2 \times 3^5 \times 5^2 \times 7^6$, $2^3 \times 5^3 \times 7^2$.

Solⁿ :- The prime numbers common to given numbers are 2, 5 & 7.

$$\therefore \text{HCF} = 2^2 \times 5 \times 7^2 = 980 \quad \underline{\text{Ans.}}$$

Ex. Find the HCF of 108, 288 and 360.

$$108 = 2^2 \times 3^3$$

$$288 = 2^5 \times 3^2$$

$$360 = 2^3 \times 5 \times 3^2$$

$$\therefore \text{H.C.F} = 2^2 \times 3^2 = 36 \quad \text{Ans.}$$

Division method :- Suppose we have to find the HCF of two given numbers. Divide the larger number by the smaller one. Now divide the divisor by the remainder. Repeat the process of dividing the preceding number by the remainder last obtained till zero is obtained as remainder. The last divisor is the required HCF.

Finding the HCF of more than two numbers : Suppose we have to find the HCF of three numbers. Then HCF of (HCF of any two) and the third number gives the HCF of three given numbers.

Similarly, the HCF of more than three numbers may be obtained.

Ex. Find the HCF of 513, 1134, and 1215.

Solⁿ:-

$$\begin{array}{r} 1134 \overline{) 1215} \quad (1 \\ -1134 \\ \hline 81 \end{array} \quad \begin{array}{r} 81 \overline{) 1134} \quad (14 \\ -81 \\ \hline 324 \\ -324 \\ \hline 0 \end{array}$$

$\therefore$ HCF of 1134 & 1215 is 81.

So required HCF = HCF of 81 & 513.

$$\begin{array}{r} 81 \overline{) 513} \quad (6 \\ -486 \\ \hline 27 \end{array} \quad \begin{array}{r} 27 \overline{) 81} \quad (3 \\ -81 \\ \hline 0 \end{array}$$

$\therefore$ HCF of given number = 27
Ans.

Ex. Reduce $\frac{391}{667}$ to lowest terms.

Solⁿ:- HCF of 391 and 667 is 23.

On dividing the numerator and denominator by 23, we get

$$\frac{391}{667} = \frac{391 \div 23}{667 \div 23} = \frac{17}{29} \quad \underline{\underline{\text{Ans.}}}$$

Least common multiple :- (LCM):-

The least number which is exactly divisible by each one of the given numbers is called their LCM.

* ① Factorization method of finding LCM →

Resolve each one of the given numbers into a product of prime factors.

Then LCM is the product of highest powers of all the factors.

② Common Division method of finding LCM :- Arrange the given numbers in a row in any order. Divide by a number which divides exactly at least two of the given numbers and carry forward the numbers which are not divisible. Repeat the above process till no two of the numbers are divisible by the same number except 1. The product of the divisors and the undivided numbers is the required LCM of the given numbers.

* Ex: Find the LCM of $2^2 \times 3^3 \times 5 \times 7^2$,

$2^3 \times 3^2 \times 5^2 \times 7^4$, $2 \times 3 \times 5^3 \times 7 \times 11$.

Solⁿ:- LCM = product of highest powers of 2, 3, 5, 7 and 11 = $2^3 \times 3^3 \times 5^3 \times 7^4 \times 11$ Ans

Ex Find the LCM of 72, 108 and 2100.

$$72 = 2^3 \times 3^2$$

$$108 = 3^3 \times 2^2$$

$$2100 = 2^2 \times 5^2 \times 3 \times 7$$

$$\therefore \text{LCM} = 2^3 \times 3^3 \times 5^2 \times 7 = 37800 \quad \underline{\text{Ans.}}$$

Ex. Find the LCM of 16, 24, 36 and 54

2	16, 24, 36, 54
2	8, 12, 18, 27
2	4, 6, 9, 27
2	2, 3, 9, 27
3	1, 3, 9, 27
3	1, 1, 3, 9
3	1, 1, 1, 3
	1, 1, 1, 1.

$$\text{L.C.M} = 2^4 \times 3^3 = 16 \times 27 = 432$$

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Ans.

* Product of two numbers = product of their HCF and LCM.

* Co-primes :- Two numbers are said to be co-primes if their HCF is 1.

* $\text{HCF} \times \text{LCM}$ of fractions :-

$$\text{HCF} = \frac{\text{HCF of numerators}}{\text{LCM of denominators}}$$

$$\text{LCM} = \frac{\text{LCM of numerators}}{\text{HCF of denominators}}$$

Ex. Find the HCF and LCM of

$$\frac{2}{3}, \frac{8}{9}, \frac{16}{81} \text{ and } \frac{10}{27}.$$

Solⁿ :-

HCF of given fractions

$$= \frac{\text{HCF of } 2, 8, 16, 10}{\text{LCM of } 3, 9, 81, 27} = \frac{2}{81}$$

LCM of given fractions

$$= \frac{\text{LCM of } 2, 8, 16, 10}{\text{HCF of } 3, 9, 81, 27} = \frac{80}{3} \quad \underline{\text{Ans.}}$$

* HCF and LCM of decimal fractions :-

On given numbers, make the same number of decimal places by annexing zeros in some numbers, if necessary. Considering these numbers without

decimal point, find HCF or LCM as the case may be. Now, in the result mark off as many decimal places as are there in each of the given numbers.

Ex. Find the HCF and LCM of 0.63, 1.05 and 2.1.

Solⁿ: $\rightarrow$ Making the same number of decimal places, the given numbers are 0.63, 1.05 and 2.10.

without decimal places, these numbers are 63, 105 and 210

Now HCF of 63, 105 and 210 is 21.

$\therefore$ HCF of 0.63, 1.05 and 2.10 is 0.21.

LCM of 63, 105 and 210 is 630

$\therefore$ LCM of 0.63, 1.05 and 2.10 is 6.30.

Ans.

* Comparision of fractions :- Find the LCM of the denominator of the given fractions. Convert each of the fractions into an equivalent fraction with LCM as the denominator, by multiplying both the numerator and denominator by the same number. The resultant fraction with the greatest numerator is the greatest.

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Q. Two numbers are in the ratio of $15 : 11$. If their HCF is 13, find the numbers.

Solⁿ:- let the required numbers be $15x$ and $11x$.

their HCF = 13

So $x = 13$

The numbers are 15×13 and 11×13

i.e. 195 and 143. Ans.

Ex. The HCF of two numbers is 11 and their LCM is 693. If one of the numbers is 77, find the other.

Solⁿ:- other number = $\frac{11 \times 693}{77} = 99$.

$\therefore$ product of numbers = HCF $\times$ LCM
Ans.

Ex. Find the least number exactly divisible by 12, 15, 20 and 27.

Solⁿ:- Required no = LCM of 12, 15, 20, 27
 $= 540 (3 \times 4 \times 5 \times 9)$ Ans.

Ex. Find the least number which when divided by 6, 7, 8, 9 and 12 leaves the same remainder 1 in each case.

Solⁿ:- Required no = (LCM of 6, 7, 8, 9, 12)

$$LCM = 3 \times 2 \times 2 \times 7 \times 2 \times 3 = 504$$

∴ required no = $504 + 1 = 505$ Ans.

Ex. Arrange the fractions $\frac{17}{18}$, $\frac{31}{36}$, $\frac{43}{45}$, $\frac{59}{60}$ in the ascending order.

Solⁿ:- LCM of 18, 36, 45, 60 = 180

Now $\frac{17}{18} = \frac{17 \times 10}{18 \times 10} = \frac{170}{180}$

$$\frac{31}{36} = \frac{31 \times 5}{36 \times 5} = \frac{155}{180}$$

$$\frac{43}{45} = \frac{43 \times 4}{45 \times 4} = \frac{172}{180}$$

$$\frac{59}{60} = \frac{59 \times 3}{60 \times 3} = \frac{177}{180}$$

Since $155 < 170 < 172 < 177$

$$\Rightarrow \frac{155}{180} < \frac{170}{180} < \frac{172}{180} < \frac{177}{180}$$

Hence $\frac{31}{36} < \frac{17}{18} < \frac{43}{45} < \frac{59}{60}$ Ans.

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(4) Simplification

① BODMAS rule :-

B $\rightarrow$ Bracket $() \rightarrow \{ \} \rightarrow []$

O $\rightarrow$ of

D $\rightarrow$ Division

M $\rightarrow$ multiplication

A $\rightarrow$ addition

S $\rightarrow$ Subtraction

② Modulus of a real number :-

$$|a| = \begin{cases} a, & a > 0 \\ -a, & a < 0 \end{cases}$$

$$|5| = 5$$

$$|-5| = -(-5) = 5$$

③ Virnaculum (or Bar) :- when an expression contains virnaculum, before applying the 'BODMAS' rule, we simplify the expression under the virnaculum.

Recurring Decimal :- If in a decimal fraction, a figure or a set of figures is repeated continuously, then such a number is called a recurring decimal.

If in a recurring decimal, if a single figure is repeated, then it is expressed by putting a dot on it.

If a set of figures is repeated, it is expressed by putting a bar on the set.

Thus

$$\frac{1}{3} = 0.333\ldots = 0.\dot{3}$$

$$\frac{22}{7} = 3.142857142857\ldots$$

$$= 3.\overline{142857}$$

Pure Recurring Decimal :- A decimal fraction in which all the figures after the decimal point are repeated, is called a pure recurring decimal.

Converting a pure Recurring Decimal into vulgar fraction : $\rightarrow$ Write the repeated figures only once in the numerator and take as many nines in the denominator as is the number of repeating figures.

$$\text{Thus } 0.\dot{5} = \frac{5}{9}$$

$$0.\overline{53} = \frac{53}{99}$$

$$0.\overline{067} = \frac{67}{999}$$

Mixed Recurring Decimal :- A decimal fraction in which some figures do not repeat and some of them are repeated, is called a mixed recurring decimal.

e.g. $0.17333\ldots = 0.17\dot{3}$

Converting a mixed Recurring Decimal into vulgar fraction $\Rightarrow$ In the numerator, take the difference between the number formed by all the digits after decimal point (taking repeated digits only once) and that formed by the digits which are not repeated.

In the denominator, take the number formed by as many nines as there are repeating digits followed by as many zeros as is the ~~numerator of~~ number of non-repeating digits.

Thus

$$0.\dot{1}\dot{6} = \frac{16-1}{90} = \frac{15}{90} = \frac{1}{6}$$

$$0.22\overline{73} = \frac{2273-22}{9900} = \frac{2251}{9900}$$

Ex: Arrange the fractions

$$\frac{5}{8}, \frac{7}{12}, \frac{13}{16}, \frac{16}{29} \text{ and } \frac{3}{4} \text{ in}$$

ascending order of magnitude.

Solⁿ: - Converting each of the given fractions into decimal form, we get

$$\frac{5}{8} = 0.625$$

$$\frac{7}{12} = 0.5833$$

$$\frac{13}{16} = 0.8125$$

$$\frac{16}{29} = 0.5517$$

$$\frac{3}{4} = 0.75$$

Now

$$0.5517 < 0.5833 < 0.625 < 0.75 < 0.8125$$

$$\Rightarrow \frac{16}{29} < \frac{7}{12} < \frac{5}{8} < \frac{3}{4} < \frac{13}{16} \quad \underline{\text{Ans.}}$$

Ex: Arrange the fractions $\frac{3}{5}, \frac{4}{7}, \frac{8}{9}, \frac{9}{11}$ in descending order.

$$\underline{\text{Solⁿ}}:- \frac{3}{5} = 0.6, \frac{4}{7} = 0.571, \frac{8}{9} = 0.88, \frac{9}{11} = 0.818$$

Now $0.88 > 0.818 > 0.6 > 0.571$

$\Rightarrow \frac{8}{9} > \frac{9}{11} > \frac{3}{5} > \frac{4}{7}$ Ans.

Ex Evaluate

(i) $6202.5 + 620.25 + 62.025$
 $+ 6.2025 + 0.62025$
 $= 6891.59775$ Ans.

(ii) $5.064 + 3.98 + 0.7036 + 7.6 + 0.3 + 2$
 $= 19.6476$ Ans.

Ex If $\frac{1}{3.718} = 0.2689$

find $\frac{1}{0.0003718}$

Solⁿ: - $\frac{1}{0.0003718} = \frac{10000}{3.718} = 10000 \times 0.2689$
 $= 2689$ Ans.

Ex. If $\frac{x}{y} = \frac{6}{5}$, find $\frac{x^2 + y^2}{x^2 - y^2}$.

Solⁿ: - $\frac{x^2 + y^2}{x^2 - y^2} = \frac{\left(\frac{x}{y}\right)^2 + 1}{\left(\frac{x}{y}\right)^2 - 1} = \frac{\frac{36}{25} + 1}{\frac{36}{25} - 1}$
 $= \frac{36 + 25}{36 - 25} = \frac{61}{11}$ Ans.

Ex. If $\frac{2x}{1 + \frac{1}{1 + \frac{x}{1-x}}} = 1$, find x .

Solⁿ:- $\frac{2x}{1 + \frac{1}{\frac{1-x+x}{1-x}}} = 1$

$\Rightarrow \frac{2x}{1 + \frac{1}{\frac{1}{1-x}}} = 1$

$\Rightarrow \frac{2x}{1 + 1-x} = 1$

$\Rightarrow \frac{2x}{2-x} = 1$

$\Rightarrow 2x = 2-x$

$\Rightarrow 3x = 2$

$\Rightarrow x = \frac{2}{3}$

Ans.

Ex(16) Find the value of

$(1 - \frac{1}{3})(1 - \frac{1}{4})(1 - \frac{1}{5}) \dots (1 - \frac{1}{100})$.

Solⁿ:- $\frac{2}{3} \cdot \frac{3}{4} \cdot \frac{4}{5} \dots \frac{99}{100}$

$= \frac{2}{100} = \frac{1}{50}$ Ans.

Ex(17) Find the value of

$\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \frac{1}{5 \times 6} + \dots + \frac{1}{9 \times 10}$

$$\begin{aligned} \text{Sol}^n: & - \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + \dots + \left(\frac{1}{9} - \frac{1}{10}\right) \\ & = \frac{1}{2} - \frac{1}{10} = \frac{5-1}{10} = \frac{4}{10} = \frac{2}{5} \quad \underline{\underline{\text{Ans.}}} \end{aligned}$$

Ex. (28) If $a^2 + b^2 = 117$ and $ab = 54$,
find $\frac{a+b}{a-b}$.

$$\begin{aligned} \text{Sol}^n: & - (a+b)^2 = a^2 + b^2 + 2ab \\ & = 117 + 2 \times 54 = 225 \end{aligned}$$

$$\Rightarrow a+b = 15 \quad \text{--- (1)}$$

$$\begin{aligned} (a-b)^2 & = a^2 + b^2 - 2ab \\ & = 117 - 2 \times 54 = 9 \end{aligned}$$

$$\Rightarrow a-b = 3 \quad \text{--- (2)}$$

$$\therefore \frac{a+b}{a-b} = \frac{15}{3} = 5 \quad \underline{\underline{\text{Ans.}}}$$

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The End